Hydrocarbons and Naming Alkanes

Formulas, IUPAC names, isomers and combustion

Name _____ Class _____ Date _____

STEMS, RULES AND DATA

Stems: meth 1 · eth 2 · prop 3 · but 4 · pent 5 · hex 6 · hept 7 · oct 8 · non 9 · dec 10

Naming: longest chain → lowest locants → methyl, ethyl… with di-/tri- → alphabetical order

Molar masses (g/mol): C 12.01 · H 1.008 · C_8H_{18} 114.22 · CO_2 44.01 · H_2O 18.02

Show units on every quantity in every step.

A. Understanding

1. Define (a) hydrocarbon (b) saturated (c) homologous series.

B. Formulas

2. Give the molecular formula of: (a) the alkane with 6 carbons (b) the alkane with 10 carbons (c) the alkene with 4 carbons (d) the alkyne with 3 carbons (e) the alkane with 20 hydrogen atoms.

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C. Naming

3. Name: (a) $CH_3CH_2CH_2CH_3$ (b) $CH_3CH(CH_3)CH_3$ (c) $CH_3CH(CH_3)CH_2CH_2CH_3$ (d) $CH_3C(CH_3)_2CH_2CH_3$ (e) $CH_3CH(CH_3)CH(CH_3)CH_3$ (f) $CH_3CH_2CH(C_2H_5)CH_2CH_3$

4. Write the condensed formula of: (a) 3-methylhexane (b) 2,2,4-trimethylpentane (c) 3-ethyl-2-methylpentane (d) but-2-ene

5. Each name is wrong. Give the correct name and say what rule was broken: (a) 1-methylbutane (b) 2-ethylpropane (c) 3-methylbutane (d) 3,3-dimethylbutane

D. Isomers and alkenes

6. Draw condensed formulas for the five structural isomers of C_6H_{14} and name each one.

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7. (a) Name $CH_2=CHCH_2CH_3$, $CH_3CH=CHCH_3$ and $CH_2=CHCH_2CH_2CH_3$. (b) Explain why "but-3-ene" is not a correct name.

E. Combustion

8. Octane, C_8H_{18}, is a component of petrol. (a) Write a balanced equation for its complete combustion. (b) Calculate the mass of CO_2 produced when 100.0 g of octane burns completely.

F. Challenge

9. A hydrocarbon burns completely to give 2.641 g of CO_2 and 1.351 g of H_2O. Its molar mass is 58.1 g/mol. (a) Find its empirical and molecular formulas. (b) Name the two structural isomers it could be.

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